**GOVT.POLYTECHNIC, PARKAL**

REPORT

ON

# SMART SIGN LANGUAGE TRANSLATOR GLOVE

Submitted in partial fulfillment of the requirement for the award of

DIPLOMA IN ELECTRONICS AND COMMUNICATION ENGINEERING

SUBMITTED BY

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**GOVT.POLYTECHNIC, PARKAL – 506164**

2023-2026

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING GOVT.POLYTECHNIC,PARKAL – 506164**

2023-2026

## CERTIFICATE

This is to certify that the project work entitled "**SIGN LANGUAGE TRANSLATOR GLOVE"** has been carried out by SHAYIZA (23204-EC-002),Partial fulfillment of the requirement of the award of DIPLOMA IN ELECTRONICS AND COMMUNICATION ENGINEERING ,By state Board Of Technical Of Education And Training Telangana During the Academic year 20232026.

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# ABSTRACT

Communication for a person who cannot hear is visual, not auditory. Generally dumb people use sign language for communication but they find difficulty in communicating with others who don't understand sign language. So there is a barrier in communication between these two communities. Smart Glove for Sign Language Translation is a work that aims to provide a cost effective system which can give to voiceless person with the help of Smart Gloves. This work consists of a glove equipped with sensors which senses different sign language gesture, these senses data are fed to Arduino UNO and transfer data to android phone via Bluetooth module, a common android phone is used in this work for sign to text and voice translation. In real life the sign language users mostly use both hands. Thus, this is a prototype work attempts to bridge the communication gap between normal ones and the disabled by designing a portable glove that captures the users gestures and outputs the translated text on a android phone. It means that using smart gloves lowers the barrier in the means of communication.

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# CHAPTER – 1
# INTRODUCTION

## DOMAIN DESCRIPTION

The number of speech impaired and hearing impaired people are increasing day by day. About 70 million people in the world are speech impaired and hearing impaired. We often come across these people communicating with the normal world. These people communicate with the help of sign language. When a speech impaired person tries to communicate with normal person and vice-versa, they feel difficult to understand. To bridge this gap, gesture recognition system is being used. Gesture recognition is a widely explored field. A lot of work has been done in the past few years. An electronic device has been used as a language interpreter and provides convenient way for communication between speech impaired, hearing impaired and normal people. Gesture recognition is classified into two main categories i.e. image processing based and sensor based. The main disadvantage of image processing based techniques includes complex algorithms for data processing. Another challenge in image and video processing includes variant lighting conditions, backgrounds and field of view constraints. The sensor based technique offers greater mobility. Smart Glove for sign language translation is our project which uses sensor based technique instead of image processing based technique. First of all, sign language is converted to analog voltage signal using accelerometer. Using Arduino UNO the code dumped into it is verified with the gesture recognized. Now, if the code matches to the gesture it transmits the data through Bluetooth module to Android phone. An application on android is used to display the characters received from Bluetooth module in the form of text and voice. The number of speech impaired and hearing- impaired people are increasing day by day. About 70 million people in the world are speech impaired and hearing impaired. We often come across these people communicating with the normal world. When a speech impaired person tries to communicate with normal person and vice-versa, they feel difficult to understand. To bridge the gap between them and normal masses, gesture recognition system is being used. Gesture recognition is classified into two main categories i.e. image processing based and sensor based. The sensor based technique offers greater mobility than the image processing based.

## AIM OF THE PROJECT

The main aim of the proposed project is to develop a cost effective system which can give voice to voiceless person with the help of Smart Gloves. It means that using Smart Glove by the disabled person enables them to communicate with others which also helps to bridge the gap between person with disability and normal person. Problems faced by the deaf person regarding employment can be overcome by this method. So in the proposed work an intelligent microcontroller based system using sensors will be developed which is able to:

- To develop code for the system when it receives its instruction from gesture recognition system using accelerometer sensors.
- To develop a micro-controller based cost effective system to recognize gesture and convert into coded form so that it can be displayed if code matches with predefined codes.
- Normal person can text their message using keyboard.

The wireless arrangement makes the device more comfortable to be used by the disabled person. Wireless transmission and reception of signals are done with the help of RF transceiver.

# CHAPTER-2

# RELATED WORK

Standalone operation and supporting expansion for future learning. over the past decade, several research studies and prototypes have been developed to assist individuals with hearing and speech impairments through gesture recognition systems. Many camera-based systems were initially proposed to capture hand gestures using image processing techniques . Although these systems achieved notable results, their accuracy heavily depended on lighting conditions, background noise, and camera position, making them less reliable in real-world conditions.

To overcome these limitations, researchers introduced wearable sensor-based solutions. Flex sensors, accelerometer, and gyroscopes have been integrated into gloves to captures finger bending and hand motion. Some works used machine learning algorithms to classify gestures into alphabets or words, while others incorporated Bluetooth and Smartphone applications for enhanced communication. However, existing solutions often suffer from high cost, lack of portability, limited gestures vocabulary, and circuit complexity.

The Smart Sign Language Translator Glove aims to overcome these limitations by designing a compact, low-power, and cost effective wearable device that translates gestures in real time into text or speech. This proposed system builds upon previous research by improving gesture detection accuracy, enabling

## 2.1 WORKING PRINCIPLE

The Smart Sign Language Translator Glove operates based on the measurement of finger bending and hand movement performed during sign language gestures. Flex sensors are attached along the fingers to detect the degree of bending. When a finger is bent, the flex sensor's resistance changes proportionally. An accelerometer is used to detect tilt orientation of the hand

These electrical signals from sensors are fed into a micro controller (Arduino UNO), which converts the analog values into digital gesture patterns. Each gesture corresponds to a predefined character or command stored in the micro-controller. Once a gesture is recognized, the system displays the interpreted output as text on a screen or serial monitor. Optionally, voice module converts this text into speech for audible communication.

Thus, the glove continuously captures hand motion and translate it into meaningful language that can be understood by individuals without sign language knowledge.

## 2.2 PRINCIPLE OF OPERATION

THE SYSTEM WORKS BASED ON THE FOLLOWING OPERATIONAL PRINCIPLES:

### 1. Introduction:

Sign language translator glove is an assistive device designed to convert hand gestures into readable or audible speech. It primarily helps people with hearing or speech disabilities communicate with others easily text.

### 2. Gesture Sensing Principle:

### The basic principle of operation is gesture sensing.

The glove contains sensors such as: •

Flex sensors placed on the fingers

- Accelerometer / Gyroscope sensors on the hand
- When a user bends fingers or moves their hand, these sensors detect the physical movement and convert it into electrical signals.

### 3. Flex Sensor Operation:

- Flex sensors change their electrical resistance when bent.

- Straight finger → low resistance
- Bent finger → high resistance
- Each finger movement produces a unique signal pattern, which helps identify different signs.

### 4. Motion Sensor Operation:

An accelerometer or gyroscope detects:

- Hand tilt
- orientation
- Up/down and left/right movements

These help identify gestures that involve full-hand motion, improving accuracy.

### 5. Signal Conversion:

- The output signals from sensors are analog in nature.
- These signals are sent to the Arduino UNO.
- The Arduino converts analog signals into digital values using its ADC (Analog-to-Digital Converter).

### 6. Gesture Processing:

- The digital sensor values are compared with predefined gesture patterns stored in the Arduino program.
- Each gesture has a unique combination of finger bends and hand movement, allowing the micro controller to recognize the correct sign.

**7. Output Generation:** After processing the gesture, the Arduino produces one of the following outputs: Text output on LCD/OLED display.

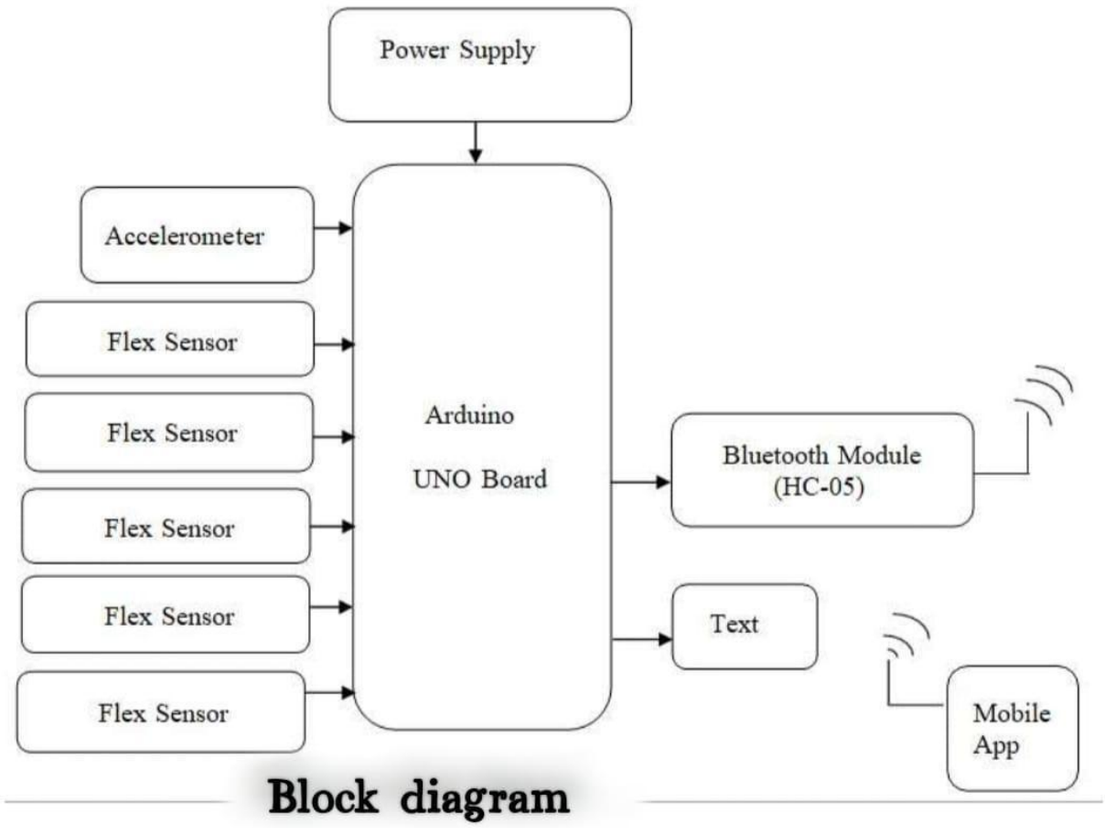
- Voice output using a speech module
- Text-to-speech output on a Smartphone via Bluetooth. This translates the hand gesture into understandable language.

### 8. Real-Time Operation:

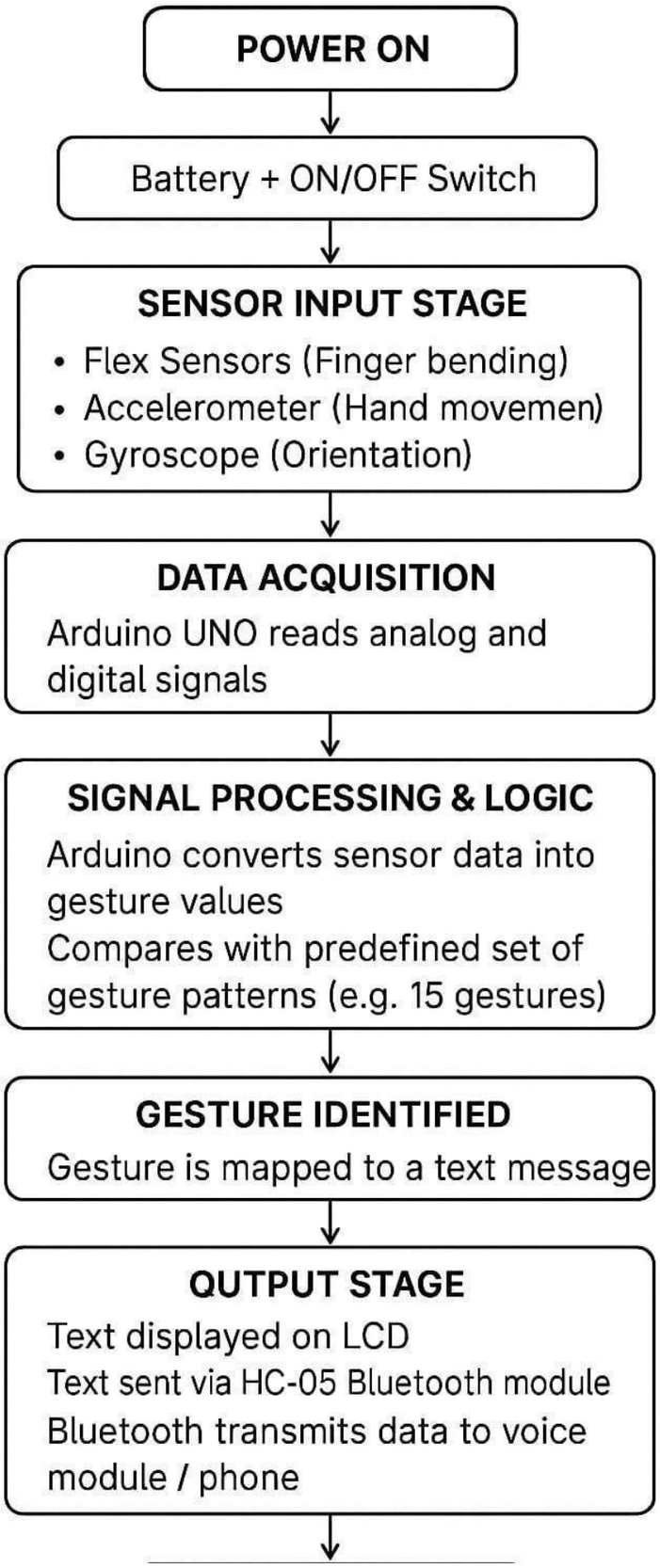
The system works continuously and instantly. As the user performs a gesture:

- Sensor detects movement.
- Arduino processes the data.
- Output is generated in real-time.

**BLOCK DIAGRAM :**



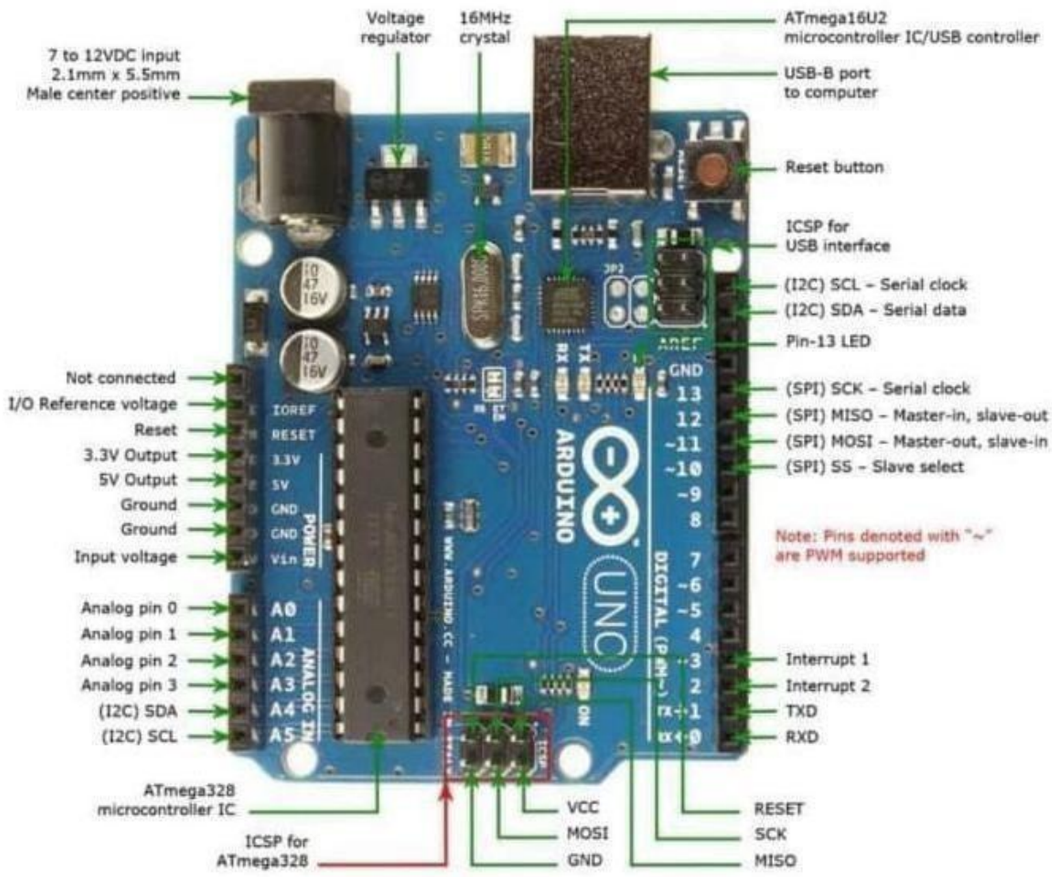

Block diagram




**FLOWCHART** **CHAPTER 3**

**COMPONENT THEORY**

**ARDUINO UNO :**




Arduino UNO is a popular open-source micro controller board based on the ATmega328P.

It is widely used in academic and hobby projects because of its simplicity, low cost, and reliability.

The board includes 14 digital I/O pins, 6 analog inputs, a 16 MHz crystal oscillator, a USB interface for programming, and a 5V regulated power system. It supports multiple communication protocols like UART, I2C, and SPI. In projects such as the Sign Language Translator Glove, the Arduino UNO acts as the main controller that reads sensor inputs, processes the data, and performs the required output actions

## FLEX SENSOR

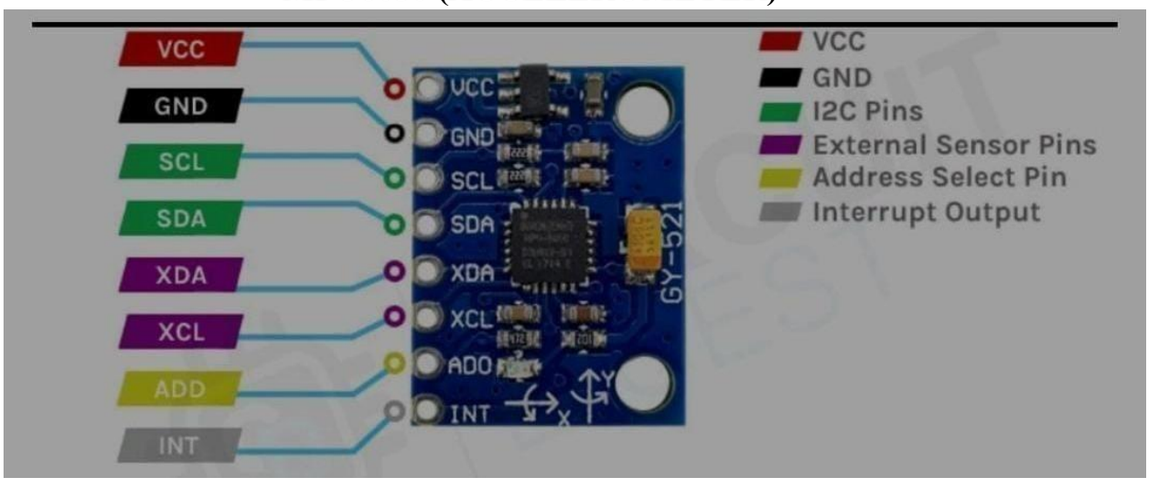


A flex sensor is a variable resistor that changes its resistance when bent. It is made of a flexible substrate coated with conductive material. As the sensor bends, the conductive path stretches, increasing the resistance. This change in resistance is read by a micro controller to measure the degree of bending. Flex sensors are commonly used in motion detection, robotics, wearable devices, and sign language translator gloves to detect finger movements accurately.

## FLEX SENSOR KEY POINTS

- Flex sensors are bend-detecting variable resistors.
- Their resistance increases as the sensor is bent.
- Made of a flexible plastic substrate with conductive ink.
- Provide an analog output that can be read by micro-controllers.
- Used to measure finger bending, joint movement, and angle changes.
- Commonly used in wearable, robotics, medical devices, and sign language gloves. Lightweight, low power, and easy to interface with Arduino.

## MPU6050 (ACCELEROMETER)




## MPU6050 - Key Points

A 6-axis motion sensor with 3-axis accelerometer + 3-axis gyroscope.

Measures tilt, linear acceleration, rotation, and orientation.

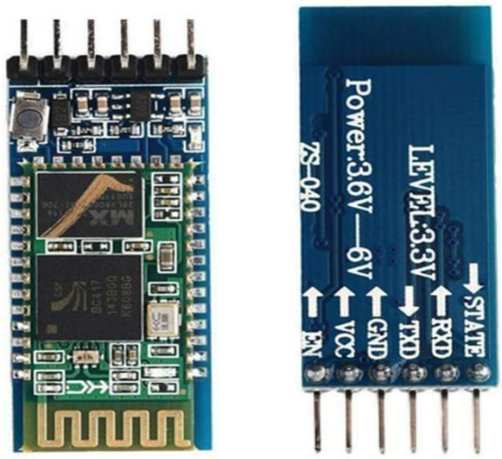
Communicates with micro controllers using the I2C (Two-Wire Interface).

Provides accurate and stable motion data for gesture recognition.

Compact, low power, and easy to interface with Arduino UNO.

Commonly used in robots, drones, gaming devices, and sign language gloves.

## HC-05 Bluetooth Module - Key Points




- A Bluetooth 2.0 module used for wireless serial communication.
- Supports UART communication with TX and RX pins.

- Can operate in Master or Slave mode.
- Works on 3.3V logic but can be powered using 5V.
- Provides a range of 10-20 meters for wireless data transfer.
- Easy to interface with Arduino, ESP32, and other micro controllers.
- Commonly used in IOT devices, wireless control systems, and sign language translator gloves.

# Boost Converter

## 1. Power Supply Stabilization

- A boost converter steps low battery voltage (like 3.7V Li-ion) up to a stable 5V or 9V.
- Ensures the micro controller (Arduino/ESP), sensors, and wireless modules operate correctly.

## 2. Required When Using Li-ion / Li-Po Battery

- Li-ion batteries drop from 4.2V → 3.0V during discharge.
- A boost converter provides constant voltage even when battery level falls.

## 3. Supports Power-Hungry Component

Used for powering:

- ESP32/ESP8266 modules
- Bluetooth modules
- Vibrating motors / haptics
- Displays (OLED, TFT)

These parts need stable 5V, which a boost converter provides.

## 4. Increases Efficiency

- Boost converters are more efficient (~85–95%) than linear regulators.
- Important for wearable devices to maximize battery life.

## 5. Compact and Lightweight

- Small size fits easily inside the wearable glove without adding bulk.

## 6. Protects Electronic Provides:

- Over current protection Short-circuit protection
- Thermal protection
- Keeps sensors and MCU safe from power fluctuations.

## 7. Common Modules Used

- MT3608 boost converter module (5V output)
- XL6009 boost converter module These are common in DIY

wearable projects .

## SPDT Switch - Key Points



- SPDT stands for Single Pole Double Throw switch.
- Has one input (common pin) and two output terminals.
- Allows selection between two different circuits or modes.
- Used to switch languages in a sign language translator glove.
- Simple, reliable, and easy to interface with Arduino digital pins.
- Provides a clear ON-ON switching action for mode selection.
- Useful in applications requiring two-state control or mode switching.

## 3.7V Lithium-ion Battery - Key points




- Rechargeable battery commonly used in portable electronic devices.
- Provides a nominal voltage of 3.7V and can reach 4.2V when fully charged.
- Offers high energy density, making it lightweight and compact.
- Provides stable and long-lasting power for projects.
- Requires a proper charging module (like TP4056) for safe charging.
- Sensitive to overcharge, over-discharge, and short circuit, so protection circuits are important.
- Suitable for wearable, IOT devices, gloves, and portable Arduino projects.

# CHAPTER – 4 CIRCUIT THEORY

## Role of Arduino UNO in the Glove

Arduino UNO functions as the brain of the entire system. It:

### 1. Collects sensor data

Flex sensors measure finger bending.

Accelerometer + gyroscope (MPU6050) measure hand tilt, movement, gestures.

### 2. Processes signals

Converts analog flex values into digital gesture decisions.

Applies thresholds and gesture-matching logic.

### 3. Controls communication

- Sends text/gesture output to the Smartphone via Bluetooth HC-05.

### 4. Manages power distribution

- Supplies 5V to sensors and modules.
- Ensures stability using onboard voltage regulators.

## Sensors/Modules Connected to Arduino UNO

**A. Flex Sensors (Finger Bending Sensors)**

- Connected to A0, A1, A2, A3, A4
- Provide analog values based on finger movement
- UNO converts analog value decision digital gesture

**B. MPU6050 Accelerometer + Gyroscope** Connected via I2C

- SDA-A4
- SCL-A5 **UNO reads:**
- Hand tilt
- Hand rotation
- Motion patterns

**C. Bluetooth Module (HC-05)**

- RX-Pin 3
- TX-Pin 2
- Powered through UNO 5V
- UNO sends final recognized word/sentence to mobile

**D. Power Components**

- 3.7V Li-ion battery Boost converter (5V) Arduino **UNO distributes power to:**
- Flex sensors MPU6050
- HC-05

**How Arduino UNO Processes Gestures** The

UNO receives sensor data continuously:

**Step 1: Flex sensor values** Example:

Index finger = 750
Middle finger = 820
Ring finger = 600

- UNO reads these using analog Read ().

**Step 2: MPU6050 orientation**

- UNO reads acceleration and gyro values.

**Step 3: Apply gesture logic**

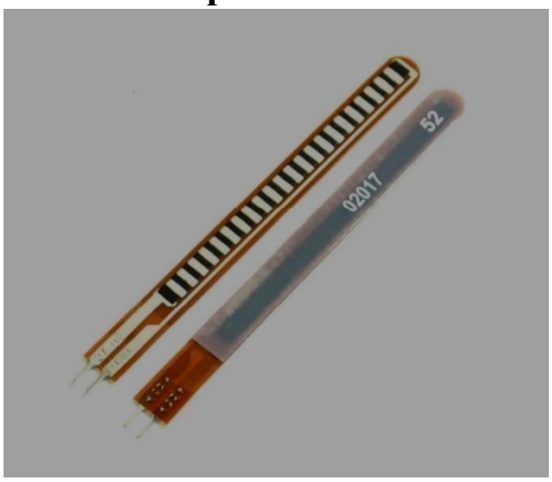
If

- Flex1 > threshold
- Flex2 low
- Hand angle = straight
- Gesture = "Hello"

**Step 4: Send output**

UNO sends "Hello" to phone via Bluetooth.

# Flex Sensor-Explanation







## 1. Introduction

A flex sensor (bend sensor) is a variable resistor that changes its resistance when it is bent. The more the sensor is bent, the higher its resistance becomes. It is commonly used to measure finger bending, wrist movement, and various motion-based inputs.

## 2. Materials Used for Manufacturing

Flex sensors are generally made using:

- Substrate material
- Plastic film (polyester or polymerize)- Flexible and durable
- Resistive element
- Carbon-based conductive ink
  (coated in a zigzags pattern on the substrate)
- Protective layer
- Laminated plastic layers to protect sensor and allow bending

Some advanced sensors may use metal ink, polymer thick film, or conductive graphite material

### 3. Specifications (General2.2-inch Flex Sensor)

- **Length**: 2.2 inches (55 mm)
- **Resistance (flat):** 10KΩ
- **Resistance (bent):** 20KΩ – 30KΩ
- **Bending angle range:** 0° to 90°
- **Power rating:** 0.5W
- **Operating voltage**: 3.3V-5V
- **Life cycle**: > 1 million bends
- **Type:** Unidirectional bend sensor (bends one direction)

### 4. How It Works

The flex sensor changes its resistance based on the bend:

**1. When straight:**

- Low resistance (around 10KΩ)

**2. When bent:**

- Conductive ink expands Resistance increases (20KΩ-30KΩ)

**3. Arduino reads analog value:**

- Higher bend- higher resistance- higher analog value
- bend -lower resistance -lower analog value

**5. Outline / Construction of Flex sensor** A typical flex sensor consists of

:

- Flexible plastic base
- Printed carbon conductive strip
- Bend-sensitive zone in the center
- Two electrical terminals for connecting wires
- Protective top laminate layer
- Adhesive layer holding the sandwich structure **Shape:**

A thin, flat, rectangular strip that bends easily.

### 6. Applications

Flex sensors are widely used in:

- Sign language translator gloves
- Robotics (joint angle measurement)
- Gaming controllers
- Virtual reality gloves
- Medical devices (rehabilitation, finger tracking)
- Wearable motion devices
- Animation & motion capture
- Industrial control systems

# 1. Introduction to MPU6050

The MPU6050 is a 6-axis motion-tracking sensor that combines:

- 3-axis accelerometer (measures linear movement)
- 3-axis gyroscope (measures rotational movement)

It is one of the most commonly used motion sensors in embedded systems. In projects like a sign language translator glove, it detects hand orientation, tilt, and motion patterns.

## 2. Specifications of MPU6050

- Accelerometer range: ±2g, ±4g, ±8g, ±16g
- Gyroscope range: ±250, 500, 1000, ±2000°/S
- Operating voltage: 3.3V 5V (depending on module)
- Communication: I2C protocol
- I2C address: 0x68 or 0x69
- Sampling rate: Up to 1 kHz
- Temperature sensor: Built-in
- Low power consumption
- Digital Motion Processor (DMP):
  - Can calculate orientation internally
  - Reduces load on microcontroller

### 3. How the MPU6050 Works

The MPU6050 contains two sensors:

**A. Accelerometer (3-axis)**

- Measures linear acceleration along X, Y, Z axes **Detects**:
  1. Tilt
  2. Movement
  3. Gravity direction

**B. Gyroscope (3-axis)**

- Measures rotation speed around X, Y, Z axes **Detects:**
  1. Hand rotation
  2. Twisting motion
  3. Angular velocity **Working Principle**

1. Accelerometer senses movement produces voltage proportional to acceleration.
2. Gyroscope senses rotation produces voltage proportional to angular velocity.
3. Internal ADC converts signals into digital values.
4. Data is sent to microcontroller using I2C (SDA & SCL pins).
5. Arduino processes the data to understand:
   - Tilt
   - Orientation
   - Gestures

## 4. Outline and each pin of MPU6050

Pin diagram &Explanation

| PIN | DESCRIPTION |
|---|---|

| | |
|---|---|
| VCC | Power supply (3.3v or 5v depending on module) |
| GND | Ground |
| SDA | 12C data line connect to ⟶ Arduino A4 |
| SCL | 12C clock line connect to ⟶ Arduino A5 |
| XDA | Aux 12C data (rarely used) |
| XCL | Aux 12C clock(rarely used) |
| AD0 | 12C address select pin (0=0*68,1=0*69) |
| INT | Interrupt pin⟶ used for alerts or DMP functions |

**5. Applications of MPU6050**

- Sign language translator glove
- Gesture recognition systems
- Robotics and drone stabilization
- Self-balancing robots (e.g., Segway)
- VR/AR hand- tracking devices
- Smartphone motion detection
- Gaming controllers Heat and body motion tracking

# 1. INTRODUCTION TO HC-05

The HC-05 is a Bluetooth module commonly used for **wireless communication** between micro controller ( like Arduino ) and Smartphone or laptops.

It supports Bluetooth 2.0 and operates in Master or Slave mode, which allows it to send and receive data over a distance of around 10 meters. In projects like a sign language translator glove, HC-05 is used to send the translated text from Arduino to a mobile app.

# 2. SPECIFICATONS OF HC-05

- **Bluetooth version**: 2.0 + EDR
- **Operating voltage**: 3.3V 5V
- **Communication:** UART Serial (TX, RX)
- **Baud rate**: Default 9600Range: 8-10 meters
- **Modes**: Master / Slave
- **Current consumption**: 30-40 mA
- **Frequency:** 2.4 GHz ISM band
- **LED indicator:** Shows pairing and communication status
- **AT command support**: Yes (for configuration )

# 3. HOW HC-05 WORKS

HC-05 communicates through serial communication (UART):

**Working Process**

1. Arduino sends data using TX/RX pins.
2. HC-05 converts serial data into Bluetooth signals.

3. Smartphone or other device receives this wireless data.
4. When paired, data is exchanged in real time like a wireless serial cable.

## Modes of Operation:

### Slave Mode:

Waits for connection. (Default mode used in Arduino projects) **Master**

**Mode:**

Connects to other Bluetooth devices.

**Pairing Indication**

- Fast blinking LED not paired ⇒ Slow blinking LED successfully paired ⇒

# OUTLINE AND EACH PIN OF HC-05

## Pin description Table

| PIN | DESCRIPTION |
|---|---|
| EN /KEY | Used to enter AT command mode |
| VCC | Power input (3.3v-5v) |
| GND | Ground |
| TXD | Transmits data from HC-05 To Arduino (connect to Arduino TX through divider if needed) |
| RXD | Receives data from Arduino (connection to Arduino TX through divider if needed) |
| STATE | Indicates connection status |
| LED | Shows power and pairing status |

**Module Outline**

- Small rectangular PCB
- Onboard LED indicator
- Bluetooth antenna printed on board Voltage regulator (allows 5V input) 6 pin header on one side:

## 5. APPLICATIONS OF HC-05

- Sign language translator gloves
- Wireless control of robots
- Home automation systems
- IoT devices
- Wireless data logging
- Bluetooth remote controllers
- Communication between Arduino boards
- Smart wearable devices
- Industrial wireless monitoring

**1. Flex Sensor Connections (5 Flex Sensors)**

Each flex sensor works with a voltage divider (using a 10k resistor).

One → 5 v end

Other → endAnalog pin + resistor to GND

Flex Sensor → Arduino Connections

| **Flex Sensor** | **Arduino Pin** |
|---|---|
| Flex Sensor 1 | A0 |
| Flex Sensor 2 | A1 |
| Flex Sensor 3 | A2 |
| Flex Sensor 4 | A3 |
| Flex Sensor 5 | A4 |

**2. MPU6050 Accelerometer / Gyroscope** Uses 12c communication.

MPU6050 PIN

| **MPU6050 PIN** | **ARDUINO UNO PIN** |
|---|---|
| VCC | 5V |
| GND | GND |
| SDA | A4 |
| INT | (optional )D2 |
| XDA | Not used |
| XCL | Not used |
| AD0 | Keep Low (for address 0*68) |

**4. Power System (Battery + Boost Converter)**

3.7V Li-ion Battery

- Connected to input of boost converter.

**Boost Converter Output (5V)**

| **Boost Converter Output** | | **Arduino Pin** |
|---|---|---|
| 5V Out + | 5V Ground | GND |

This power:

→ Arduino UNO

→ Flex sensors

→ MPU6050

→ HC-05

All grounds must be common.

**5. Optional Components**

**SPDT Switch**

- Input: Battery → Output:
    a. "ON" Boost converter
    b. "OFF" Disconnected **LED Indicator**

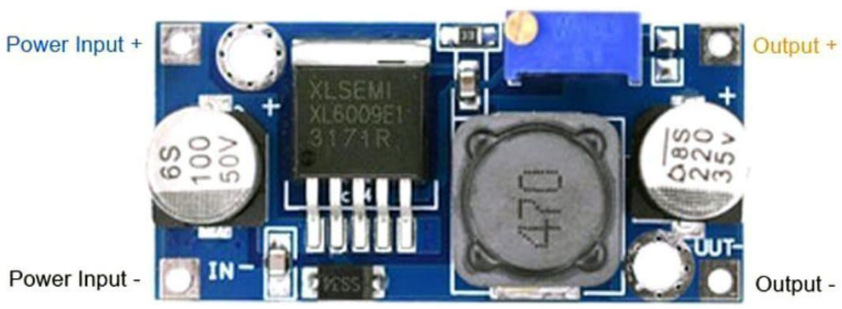
LED → +D13

LED - → GND

**Full Wiring Summary (Very Short for Documentation)**

- Flex Sensors → A0, A1, A2, A3, A4
- MPU605 → SDA-A4, SCL-A5, VCC-5V, GND-GND
- HC-05 TX → D2, RX → D3 (voltage divided), VCC → 5V, GND → GND
- Battery Boost Converter → Arduino 5V

**BUCK STEP UP CONVERTER**




## 1. Introduction

A Buck-Boost Converter is a DC-DC power converter that can increase (step-up) or decrease (step-down) the input voltage to a required output voltage.

It uses switching components (inductor, MOSFET/transistor, diode, capacitor) to regulate voltage efficiently.

It is widely used in battery-powered systems where the input voltage keeps changing.

**2. Specifications (General Buck-Boost Converter Specs)** Different modules vary, but common specifications include:

**Input Voltage Range:**

Usually 2V to 24V (varies by module) **Output Voltage Range:**

Can be higher or lower than input, like 5V, 9V, 12V, 24V **Output Current:**

1A to 3A typical (depending on module) **Efficiency**:

Up to 85-95%

**Switching Frequency:**

Typically 150 kHz - 1 MHz

**Protection Features (optional):**

- Over-current protection
- Over-temperature protection
- Short-circuit protection
- Reverse polarity protection

## 3. How It Works

A Buck-Boost converter controls energy flow using switching:

**Step-by-Step Working**

**(i) Switch ON (Charging Phase)**

- MOSFET turns ON
- Inductor stores energy from the input battery
- Output is isolated from the input during this time **(ii) Switch OFF (Discharging Phase)**

- MOSFET turns OFF
- Inductor releases stored energy
- Through the diode, the energy reaches the output capacitor
- Output voltage increases to a level set by the controller

**Voltage Control**

The converter adjusts the duty cycle (ratio of ON/OFF time) to maintain constant output voltage.

- Higher duty cycle Boost (Step-Up)
- Lower duty cycle Buck (Step-Down)

## 4. Applications

Buck-Boost converters are used everywhere in modern electronics, especially where stable voltage is needed from unstable sources. **Common Applications**

- Battery-powered devices (voltage drops as battery discharges)
- Power banks
- LED drivers
- Solar energy systems
- Arduino/ESP32/Raspberry Pi power supply
- Automotive electronics
- Portable medical electronics
- Robotics and IOT devices

# CHAPTER 5

## ADVANTAGES AND LIMITATIONS

## 5.1 ADVANTAGES

➤ Any complications in its user interface would inhibit the glove's use in everyday life. The user can begin translation without much difficulty. Each translation can be done without any unnecessary button-pressing or other interfacing.

It has very few components and is cost effective. It is affordable by any kind of person who needs it.

➤ For marketability purposes, the device shall be aesthetically pleasing and easily wearable without any factors that hinder convenience during extended periods of time. This includes a smooth, professional appearance without any components that irritate, cut, bruise, or otherwise cause discomfort for the user.

➤ It is small in size hence, it is flexible to users. It is easy to operate and is a real time translation.

### DISADVANTAGES

The main disadvantage is, as it uses Bluetooth module it ranges up to 100meters only.

➤ It is delicate as it has wired connections on it. Glove must be handled with care without any loss of connections.

### APPLICATIONS

➤ It can be used in hospitals that need several measurement systems which can investigate physiological parameters of the patients.

It can act as a communication aid for the hearing and speech-impaired persons.

➤ Using this we can control any electronic machines and devices wirelessly by adding some others gestures and commands.

This project can be also used as Signed Language Convertor/Interpreter.

➤For gaming purposes like 3D gaming, at the place of joysticks.

It can be used in military actions based on hand gestures, which can be used for squad communication.

➤ The hand gesture recognition system can be used in robotics, desktop and tablet PC applications and gaming.

## 5.2 LIMITATION Limitations of Sign Language Translator Glove

1. **Limited Vocabulary**
   - The glove can recognize only the signs that are programmed.
   - It cannot understand all complex gestures used in full sign language.
2. **Difficulty Detecting Finger Combinations**
   - Some signs require multiple finger bends and hand shapes.
   - Simple flex sensors may not capture every small movement accurately.
3. **Not Accurate for Fast Movements**
   - If the user signs quickly, the sensors may not read properly. This can cause incorrect translation.
4. **Depends on Proper Glove Fit**
   - If the glove is loose or too tight, readings from sensors become unreliable.
5. **Limited Gesture Types**
   - Many gloves detect only static gestures.
   - Dynamic gestures (movement-based signs) may not be recognized without accelerometer/gyroscope
6. **Power and Hardware Limitations**
   - Arduino Requires battery or continuous power supply.
   - -based systems may lag or have limited processing capability.
7. **Environmental Interference**
   - Temperature, sweat, and dust can affect sensor accuracy. Flex sensors may degrade over time.
8. **Not a Complete Communication Tool**
   - It translates signs but does not interpret facial expressions or body language, which are important in sign language.
9. **User Skill Required**
   - The user must sign correctly for accurate translation. Beginners may struggle with exact gestures.
10. **Prototype Stage**
    - Most translator gloves are experimental and not ready for commercial use. Durability and long –term performance are not guaranteed.

# CHAPTER 6
## FUTURE DEVELOPMENT AND CONCLUSION

### 6.1 FUTURE WORK

Thus the gesture recognition system designed using sensor fusion and gesture recognition techniques in this venture has a lot of future aspects that has to be taken into consideration in order to support the help for this differently able people more. This smart glove readily banishes the required interpretation between a speech impaired and a normal person.

Future implementation can be made by enhancing the quality of the mobile application which can be used to produce a lot of technical quality research as in what is to be implemented to assist them more. We developed for Android OS; In future we can also develop an application so that it can work on any platform. We can also reduce the number of connecting wires to make it simple to wear. It can be implemented in various fields like in airport and railway stations to assist the speech impaired. One more technical issue can be handled is to assist multi gesture at a higher speed in which at times this device accuracy fails to reach the peak. Keeping in mind the end goal to enhance and encourage the more signal acknowledgment, movement handling unit can be introduced.

### 6.2 CONCLUSION

Sign language is one of the useful tools to ease the communication between the deaf and mute communities and normal society. Though sign language can be implemented to communicate, the target person must have an idea of the sign language which is not possible always. Hence our project lowers such barriers .The glove is capable of translating their sign language gestures into speech through android phone. Smart glove focuses the translation of gestures of the alphabets. Comparing with other approaches, smart glove uses Principle Component Analysis to classify the real time input data for feature extraction.

Disabled use these gloves to convert sign performed by them into speech and text. This paper is a useful tool for speech impaired and partially paralyzed patients which fill the communication gap between patients, doctors and relatives.

- From the convenience of simple accelerometer, a user is able to interact with others in more comfortable and easier manner.
- This makes it possible for the user to not only interact with their community but with others also and they can also live normal life.
- It is portable to use and is cost effective.
- This paper will give dumb a voice to speak for their needs and to express their gestures
- Hence this paper is an attempt to make it easy to understand the actions of the dumb people by getting the output in the form of text and voice.
- The text is also forwarded as SMS via Bluetooth or modem for better convenience and for security purposes.
- The end product will have a cheap and simplistic design making it easy for users to interact easily.
- The system is capable of recognizing signs more quickly. Further more real time recognition ratio of nearly 99% can be easily achieved.

# CHAPTER 7 CODE:

```
/**********************************
                Sign Language Translator Glove Updated
                Version:
```

- MPU6050: SCL → A5, SDA → A4
- Flex sensors: A0, A1, A2, A3
- HC-05 Bluetooth: TX → D2, RX → D3
- Language switch: D7

```
                        LOW = English  HIGH
                        = Telugu
**********************************/

                        #include <Wire.h>
                    #include <Arduino_MPU6050.h>
                    #include <SoftwareSerial.h>

                        MPU6050 mpu;
                    SoftwareSerial BT(2, 3); // RX, TX

                        // Flex sensor pins
                    #define adc1 A0  #define
                    adc2 A1  #define adc3
                    A2  #define adc4 A3

                        // Language switch pin
                    #define langSwitch 7

                int flex1 = 0, flex2 = 0, flex3 = 0, flex4 = 0;

                        void setup()
                        {
pinMode (langSwitch, INPUT_PULLUP); // HIGH = Telugu, LOW = English
Serial.begin(9600);
```

```
BT.begin(9600);
Wire.begin();
Void loop()
{ // Read flex
sensor values flex1 =
analogRead(adc1); flex2
= analogRead(adc2);
flex3 =
analogRead(adc3); flex4
= analogRead(adc4);
delay(10);

// Update MPU6050 data
mpu.update();

float normX = mpu.getAccelerationX();
float normY = mpu.getAccelerationY();
float normZ = mpu.getAccelerationZ();

delay(200);

String message = "";

// -------- Gesture Logic -------if
((flex2 > 950) && (flex1 > 940))
{ if (normX > 5)
message = "YES";   else if (normX
< -3) message = "NO"; else if (normY
> 3) message = "OK";   else if (normY
< -3) message = "BYE";
} else if
(flex1 > 950)
{ if (normX > 5)
message = "THANK YOU"; else if (normX < -5) message =
"NICE TO MEET YOU"; else if (normY > 5) message =
"GLAD TO BE HERE"; else if (normY < -5) message =
```

```
"HELLO — this is my project"; } else if (flex2 <
950)
{
if (normX > 5) message = "WATER"; else if
(normX < -5) message = "WHAT IS YOUR NAME"; else if
(normY > 5) message = "SORRY"; else if (normY < -5)
message = "FINE";
}
else
{
if (normX > 5) message = "HELP"; else if
(normX < -5) message = "WASHROOM"; else
if (normY > 5) message = "MEDICINE"; else if
(normY < -5) message = "FOOD";
}

// ------ Language Selection ------
int lang = digitalRead(langSwitch); // HIGH = Telugu, LOW = English

String output = "";

if (message != "")
{ if (lang
== LOW)
{
// ----- English ----output
= message;
}
else
{
// ----- Telugu Translations ----- if
(message == "YES") output = "అవు";
else if (message == "NO") output =
"కా"; else if
```

(message == "OK") output = "సర"; else if

(message == "BYE") output = "వీడ్";

else if (message == "THANK YOU") output = "ధన్య వ్"; else if

(message == "NICE TO MEET YOU") output = "మిమ్మ కలవడం ఆన్ందం";

else if (message == "GLAD TO BE HERE") output =

"ఇకడడ ఉండటం

ఎంత సంతం";

else if (message == "HELLO — this is my project") output = "హల — ఇద న్
ప్రా";జ

else if (message == "WATER") output = "నీళ";

else if (message == "WHAT IS YOUR NAME") output = "మ్ పే ఏమి"; else if

(message == "SORRY") output = "కమ్ంమ్ం"; else if (message ==

"FINE") output = "బాన్మ మ్ ";

else if (message == "HELP") output = "సహం చం"; else if

(message == "WASHROOM") output = "బపతమ్"; else if

(message == "MEDICINE") output = "ోధం"; else if

(message == "FOOD") output = "ఆహం";

}

// Send final output

Serial.println(output);

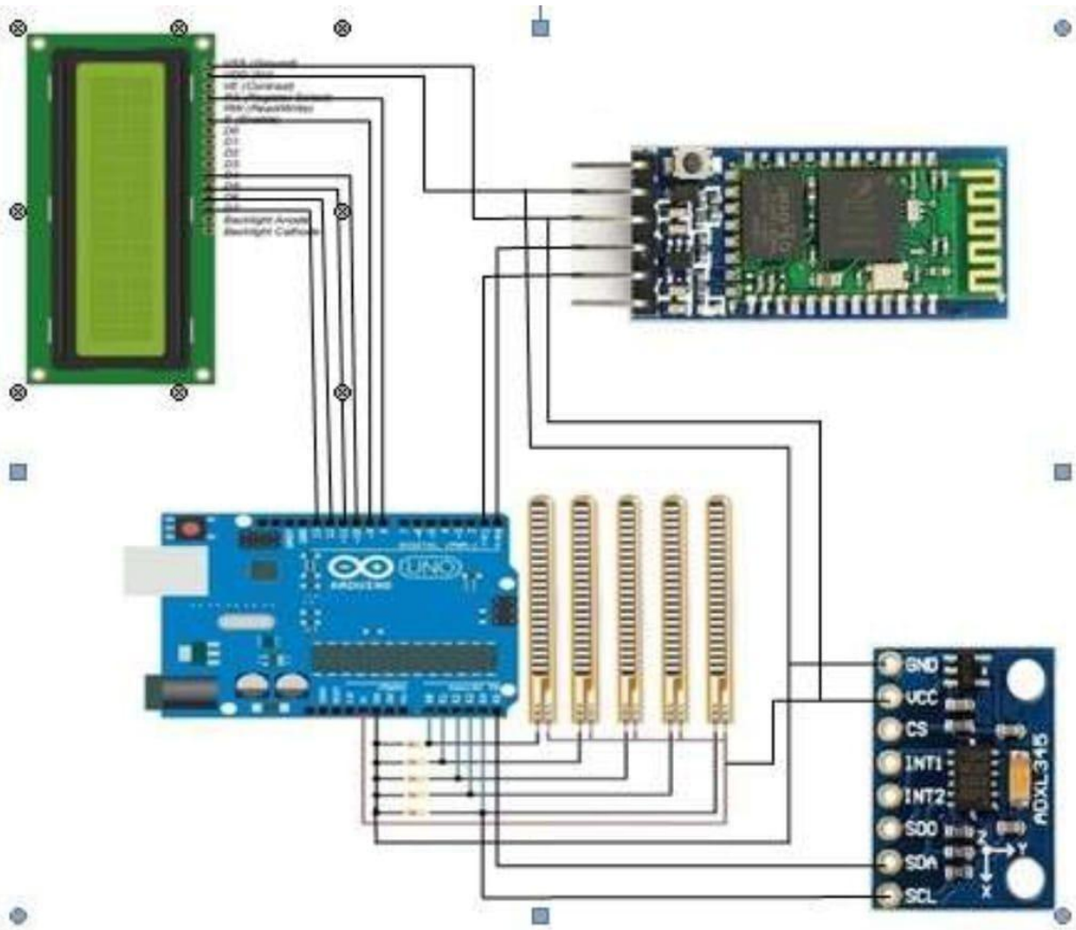
BT.println(output);

}

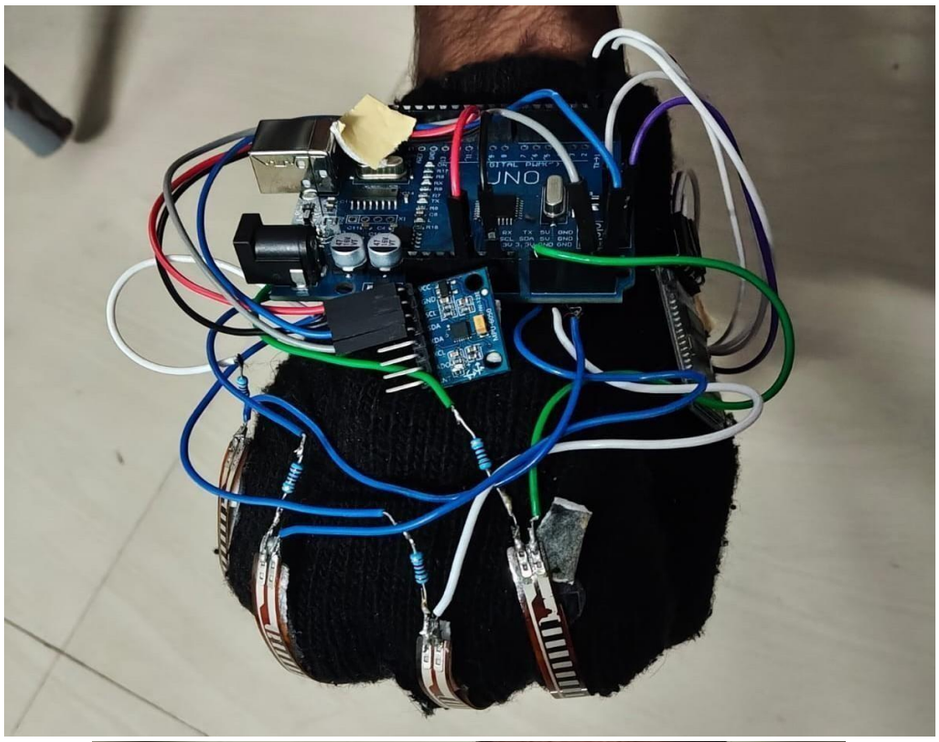
delay(300);

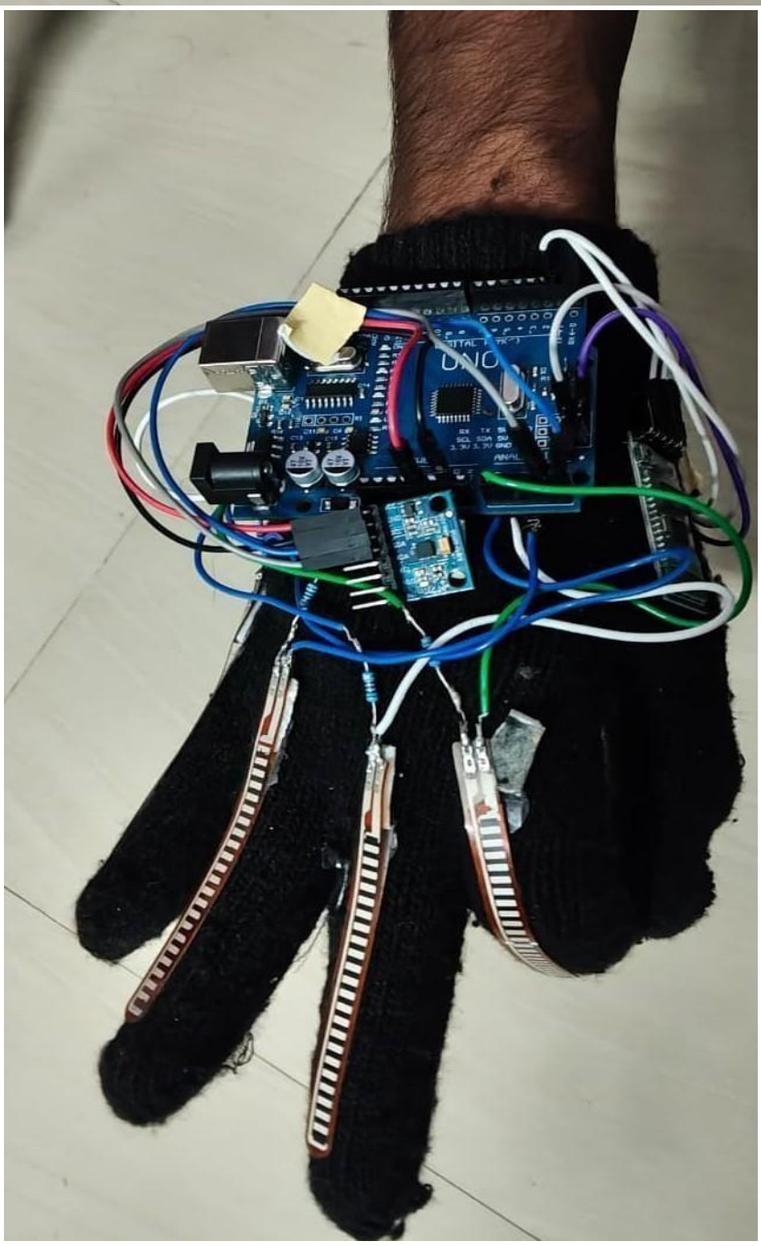
}

# 7.1 CIRCUIT DIAGRAM

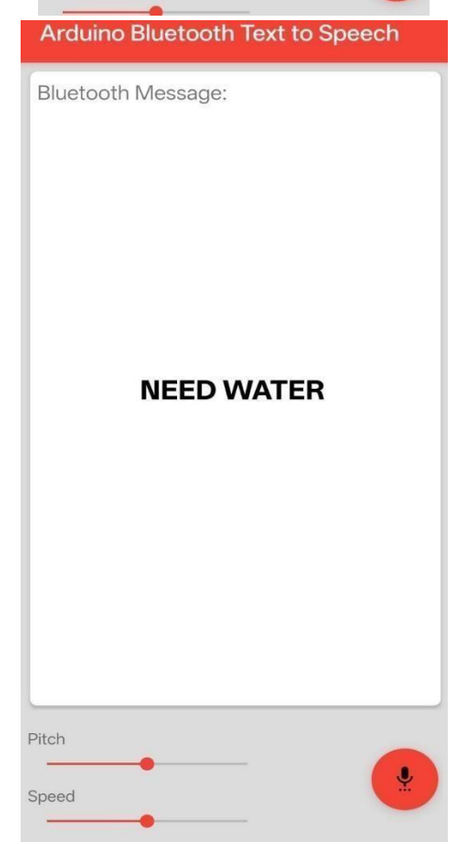



## 7.2 OUTPUT






Arduino Bluetooth Text to Speech
Bluetooth Message:
HELLO
Pitch
Speed


Arduino Bluetooth Text to Speech
Bluetooth Message:
NEED WATER
Pitch
Speed




**THANKS**

# CHAPTER 8

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